

# Naga Pranav Puppala

Coppell, TX | 945-345-1322 | [pranav.puppala0206@gmail.com](mailto:pranav.puppala0206@gmail.com) | [pranav-puppala.vercel.app](https://pranav-puppala.vercel.app) |  
[linkedin.com/in/pranav-puppala](https://linkedin.com/in/pranav-puppala) | [github.com/PranavPuppala](https://github.com/PranavPuppala)

## EDUCATION

**The University of Texas at Dallas**

*Master of Science, Computer Science - Intelligent Systems*

Richardson, TX

Aug. 2026 – Dec 2027

**The University of Texas at Dallas**

*Bachelor of Science, Computer Science*

Richardson, TX

Jan. 2023 – May 2026

## EXPERIENCE

**Software Engineer Intern (Capstone Project)**

Feb 2026 – May 2026

*UT Southwestern Medical Center*

*Dallas, TX*

- Built a **hybrid RAG pipeline** using **LangChain**, **ChromaDB**, **BM25**, and **cross-encoder reranking** to generate clinically accurate, persona-aware **Major Depressive Disorder** patient responses for psychiatric interview training
- Designed a **synthetic persona generation pipeline** that extracted structured clinical profiles from therapy transcripts using **Groq** and **LLaMA**, scored multi-dimensional patient similarity via **ChromaDB**, and synthesized blended **MDD personas** for real-time conversational simulation
- Developed a **real-time psychiatric patient simulation system** with structured **behavioral metadata tagging** (mood, affect, and symptom) producing both natural dialogue and **clinical JSON output**

**Undergraduate Research Assistant**

Feb 2024 – May 2025

*The University of Texas at Dallas*

*Richardson, TX*

- Developed **network intrusion detection systems** using machine learning techniques including **RandomForest**, **EllipticEnvelope**, and **regression models**
- Trained machine learning models to identify **network traffic anomalies** focusing on **SYN flood** and **DDoS attack detection**
- Reduced false positive rates by **30%** and achieved **92% precision** and **88% recall** using one-class classifiers with unsupervised machine learning in **Python**

## PROJECTS

**ExpenseTracker** | *Django REST, React, JWT Authentication*

Aug 2025 – Oct 2025

- Built **full-stack expense management application** with **Django REST API** backend featuring **JWT authentication**, user registration, login, and **CRUD operations** for expense tracking
- Implemented **filtering and pagination systems** with **real-time search**, category-based filtering, and **dashboard analytics** with trend visualization using **Recharts**
- Designed UI with **React** and **Tailwind CSS** including dark theme design, form validation, and user profile management with password change functionality

**Resume Analyzer** | *React, TypeScript, Puter.js*

Jul 2025 – Aug 2025

- Created **web-based resume analysis platform** integrating **Puter.js** for user authentication, file uploads, **AI feedback**, and persistent key-value storage
- Designed **end-to-end flow** where users upload resumes, receive structured **ATS evaluations from Claude**, and view personalized feedback in dynamic UI
- Created custom **React components** including interactive score gauges, tip accordions, and resume preview cards to enhance user experience

**E-commerce website** | *React, PostgreSQL, Prisma*

Sep 2024 – Nov 2024

- Developed **search and filter features** with **FuseJS** for fuzzy product search and category filtering
- Used **Prisma** with **PostgreSQL** to implement efficient product queries and dynamic page generation

## TECHNICAL SKILLS

**Languages:** Java, Python, C++, SQL, PostgreSQL, JavaScript, HTML, CSS

**Frameworks:** React, Next.js, Django, Scikit-Learn, Tensorflow

**Developer Tools:** Git, VS Code, Github, Claude, ChatGPT, Perplexity AI, Github copilot

**Libraries:** Pandas, NumPy, Matplotlib, Axios, Prisma, Tailwind CSS, Recharts, Zustand